

SAM.GOV ANNOUNCEMENT TYPE: SPECIAL NOTICE

ANNOUNCEMENT NUMBER: ACQ0042379

RELATED NOTICE: NIST-SS25-CHIPS-0021

SUBJECT: NOTICE OF INTENT TO SOLE SOURCE for Test Vehicle to Study Reliability of Materials Used in Advanced Packaging of Semiconductor Chips

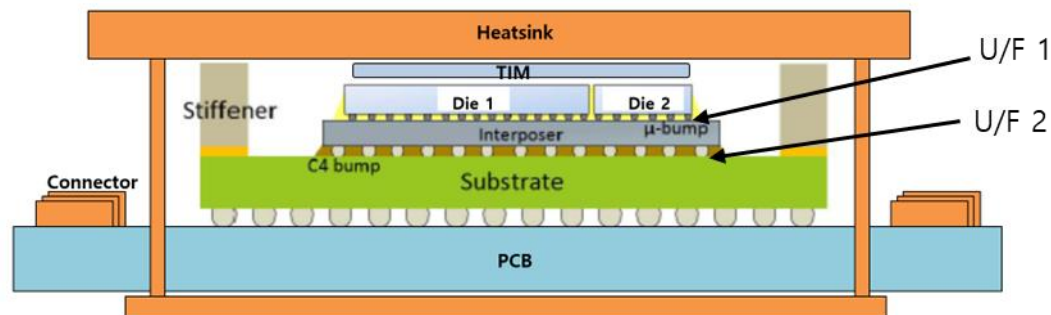
Notice of Intent to Noncompetitively acquire upgrade a Test Vehicle to Study Reliability of Materials Used in Advanced Packaging of Semiconductor Chips

This notice is not a request for a quotation. A solicitation document will not be issued, and quotations will not be requested.

This acquisition is being conducted under the authority of FAR 13.106-1(b). The North American Industry Classification System (NAICS) code for this acquisition is 334516, Analytical Laboratory Instrument Manufacturing which has a small business size standard of 1,000 employees.

In support of the CHIPS Metrology program, reliability testing and modeling of polymeric components in advanced packaging for semiconductors will be studied and developed. These materials will include underfill epoxies, thermal interface materials, and redistribution layer components. The long-term aging behavior of these materials will be studied using accelerated aging experiments. To study interfacial material properties, material interactions, and failure modes, advanced test vehicles that include all three material components and advanced geometries in a realistic heterogeneously integrated package are required. These test vehicles will be monitored in-situ during aging experiments and studied throughout accelerated long-term aging.

The diagram below shows a representative test vehicle for achieving the goals of the study.



NIST requires a quantity of 200 Advanced Test Vehicles that meet all technical specifications identified below.

- a. Design
 - i. Schematic Design Capabilities
 1. Develop schematics for the Logic (Die 1) and Memory (Die 2) die that contains daisy chain circuits, heaters, and temperature sensors.

2. The entire schematic should be appropriately marked for critical routing and other significant CAD layout information.
3. Specify EDA vendor tool along with any special terms and conditions or exceptions.
- ii. Layout Capabilities
 1. Layout and route the design from the generated schematics, considering factors such as limited number of utilizable I/O of μ bump, C4, and BGA to the socket in PCB.
 2. Layout should be DRC clean per established foundry Design Rules and match the schematic with a LVS checker
 3. Generate netlist from μ bump, C4, BGA, and sockets for proper testing features.
 4. Must include integrated resistivity-based sensors throughout test vehicle structure for *in-situ* monitoring and heating. Number and details to be defined by NIST scientist and service provider (approximately 300-500 expected).
- iii. Components
 1. Die
 - a. Logic (Die 1) and HBM (Die 2) die to be used for chiplet monitoring
 - b. 2-3 layers of build up (for daisy chain, temperature sensors, heating).
 2. Silicon Interposer
 - a. Design and fabricate interposer to connect and route signals between the 2 die with RDLs and TSVs.
 - b. Ability to connect the interposer to the die with μ bumps on top and to the substrate with C4 bumps on the bottom.
 - c. Application of underfill for μ bump and C4 bump layers
 - d. Design and place test features such as daisy chain for μ bump and C4 as well as EM and fatigue for RDL and via.
 3. Substrate
 - a. Design High Density Organic Substrate to mount interposer
 4. Integrated resistivity-based sensors
 5. TIM1
 6. Connector
 7. Stiffener
 8. Heatsink with clamp
- iv. Variability
 1. TIM1 coverage and material conductivities
 2. Interposer to substrate ratio
 3. Lid to chip ratio
 4. RDL line width, thickness, number of layers
 5. μ bump pad diameter and pitch
 6. TSV diameter: aspect ratio
 7. Flipchip (C4) and μ bump underfill selection
 8. Stiffener
- v. Modularity
 1. Chiplet and package can include different combinations of designs and sizes (TSV, μ bump, and C4 bumps) and materials (RDLs, stiffener, TIM1s, and/or underfills) similar to a Design of Experiments (DoE) which would be defined by NIST scientist and service provider.
- b. Fabrication and Assembly
 - i. Fabrication Process

1. Tapeout
 - a. Generate mask order at the foundry and create reticle set.
2. Wafer Fabrication
 - a. Run at least 3 wafers through the wafer fabrication process in the foundry.
- ii. Assembly
 1. Assemble the entire system using necessary bumps and then mount on PCB as shown in the figure, including stiffener.
 2. Connectors as shown in figure.
 3. Heatsink with clamp as shown in figure.
- iii. Test
 1. Test vehicles must undergo testing to ensure quality control, electrical connectivity, and reliability. Details will be defined by NIST scientist and service provider.
- c. Application
 - i. Primary function is to characterize failure mechanisms during long-term aging to model reliability in packaging materials such as RDL, TIM1s, and Underfills.
 - ii. Additionally, this test vehicle would be used for the following purposes:
 1. Thermo-Mechanical Warpage Assessment – 3D DIC *in-situ* Deformation Measurement
 2. Thermal Performance Characterization – TIM1 Degradation
 3. Fatigue Life Assessment of μ bump, C4, RDL Trace, and Via in Si Interposer.
 4. TSV Failure – Fatigue, Creeping, Si Cracking, Delamination.
 5. Electromigration of RDL/Via in the Interposer

NIST conducted market research in November 2024 to March 2025 to determine what sources could potentially meet NIST's minimum requirements. This included searches on GSA; SAM; surveying vendor websites; and issuance of a Sources Sought Notice on SAM.gov. Market research identified a various entities ranging from private companies, Universities, and national laboratories that could potentially meet requirements, but only one can provide a commercial product that combines underfill epoxies, thermal interface materials, and redistribution layer components along with advanced geometries in a realistic heterogeneously integrated package including required embedded sensors and memory and logic die. The results of market research revealed that only The Research Foundation for The State University of New York (UEI: NQMVAQUFU53) appears to be capable of meeting NIST's requirements.

However, any sources that believe they are capable of meeting NIST's minimum requirements are encouraged to respond to this notice by the response date to provide the following information at a minimum: Company Unique Entity Identifier (UEI) number in <https://sam.gov>; details about what your company is capable of providing that meets or exceeds NIST's minimum requirements; whether your company is an authorized reseller of the product or service being cited and evidence of such authorization; and any other information that can help NIST determine whether this requirement may be competitively satisfied.

A determination by the Government not to compete the proposed acquisition based upon responses to this notice is solely within the discretion of the Government. Information received will be considered solely for the purpose of determining whether to conduct a competitive procurement. Only responses received by the offers due date and time of this notice will be considered by the government. Responses shall be submitted via email to nina.lin@nist.gov.